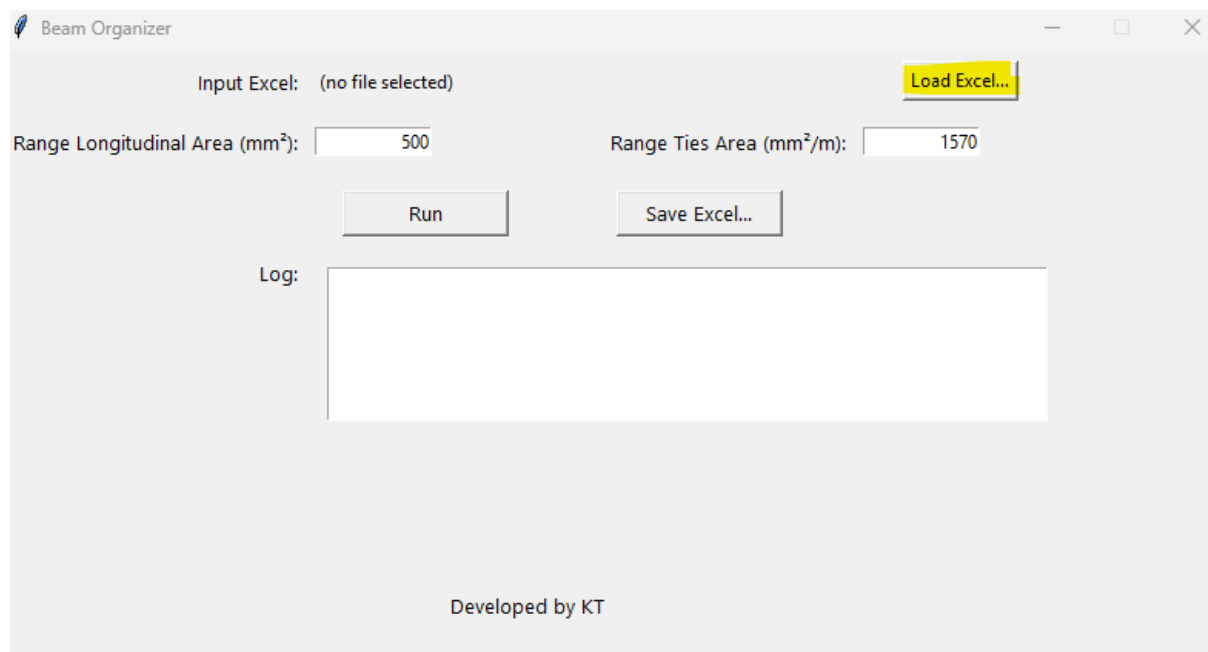
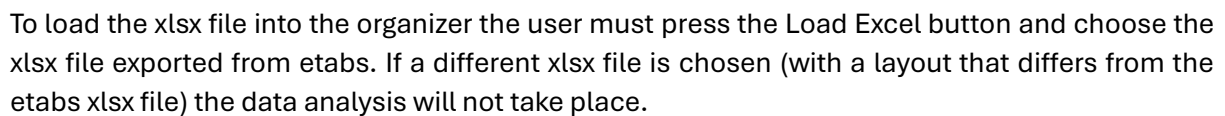


This software organizes the beam reinforcement into groups. After performing the analysis using Etabs, the user must extract the Conc Bm Sum – Eurocode 2-2004 and save it as an .xlsx file.



Once the file is loaded the program performs the following data analysis:

1. Etabs provides reinforcement area for various cross sections across the length of the beam. The program finds out the maximum required reinforcement area (top, bottom and ties) across the element for every member. This information is displayed at the Unique Names Reinforcement sheet of the output excel file. This sheet also displays the unique name of each member (this name is assigned by etabs) and the cross-section name.

	A	B	C	D	E	F
	DesignSect	UniqueName	As Top Total (mm ²)	As Bottom Total (mm ²)	As Transverse Total (mm ² /m)	
	Beam 30x50	100	712	718	955.05	
	Beam 30x60	1001	1548	1233	2081.01	
	Beam 30x60	1002	1976	909	2236.01	
	Beam 30x60	1003	1293	760	1383.41	
	Beam 30x60	1004	4004	2427	3025.93	
	Beam 30x60	1005	3818	2233	3315.36	
	Beam 30x60	1006	3048	2287	3299.2	
	Beam 30x60	1007	3909	2247	3929.32	
	Beam 30x60	1008	2282	1507	2173.34	
	Beam 30x60	1009	3212	1686	2653.55	
	Beam 30x50	101	1590	973	1862.48	
	Beam 30x60	1010	1413	760	1387.61	
	Beam 40x60	1011	5135	2792	4554.73	
	Beam 30x60	1012	3115	2066	3160.98	
	Beam 30x60	1013	3581	2109	2797.17	
	Beam 80x70	1014	3629	2402	3983.79	
	Beam 30x60	1015	1186	760	1666.69	
	Beam 30x60	1016	1755	930	2175.96	
	Beam 30x60	1017	4023	2577	4350.31	
	Beam 30x60	1018	2997	1768	3088.35	
	Beam 30x50	1019	1579	761	1210.11	
	Beam 30x60 Cantilever	102	1408	760	2469.24	

2. Design Section Summary displays the absolute maximum reinforcement for each different member type. For every different member type the maximum top, bottom and ties reinforcement is displayed. Also, the mean (average), standard deviation and coefficient of variation is displayed for these values.

DesignSect	As Top Total (mm ²)	As Bottom Total (mm ²)	As Transverse Total (mm ² /m)	As Top Mean (mm ²)	As Bottom Mean (mm ²)	As Bottom Std (mm ²)	As Transverse Mean (mm ² /m)	As Transverse Std (mm ² /m)	AsTopCOV[%]	AsBotCOV[%]	AttrnCOV[%]	
Beam 25x25	511	310	1674.59	493.5	24.74873734	297	18.38477631	1413.68	368.9824606	5.014941711	6.19016037	26.1008475
Beam 30x50	3711	2807	5621.75	1497.05	606.8476373	1080.568056	431.8737174	1477.421639	663.4595339	40.53634323	39.9672853	44.9065803
Beam 30x60	4835	3612	5984.64	2257.35	1071.854832	1601.70997	703.6456584	2203.185166	1047.020901	47.4828087	43.9309033	47.5230551
Beam 30x60 Cantilever	4142	2275	7333.22	2606.63	855.5799989	1527.263158	430.8780237	5589.769474	1189.96986	32.82320393	28.2124283	21.288353
Beam 30x70	2584	2501	4718.05	2584		2501		4718.05				
Beam 40x50	3760	2499	3217.63	1672.14	682.63253	1300.727273	597.275017	2050.714091	619.3194186	40.82397494	45.9185434	30.2001835
Beam 40x60	6194	4052	5785.75	4366.15	1055.189396	2770.8	655.9247068	3499.692	1318.955181	24.16750216	23.6727534	37.6877503
Beam 50x50	6072	4430	6706.98	4211.5	2631.144333	2990	2036.46753	5965.965	1047.953463	62.47523051	68.1092819	17.5655315
Beam 50x60	7771	4848	8972.64	3763.51	1651.327647	2493.818898	945.5008954	3622.644331	2173.942025	43.87730742	37.9137754	60.0098112
Beam 50x70	9486	5526	6295.63	8169.5	945.931111	4762	936.1748412	5907.42	374.1837238	11.57881279	19.6592785	6.33413104
Beam 70x60	9030	4862	5887.54	6684	3317.745017	4681.5	255.265548	5644.595	343.5761139	49.63711875	5.45264441	6.08681604
Beam 80x70	6533	10961	15805.8	5081	2053.438093	6681.5	6052.12694	9894.78	8359.402225	40.41405417	90.5803628	84.4829519
Coupling Beam 60x100	6069	6005	10121.2	4319.31	1320.774102	4296.75	1272.330644	7730.159375	1765.239051	30.57834093	29.6114655	22.8357394
Coupling Beam 60x60	8135	7821	7758.83	4504.94	1902.584664	4400.926829	1735.024828	4549.559268	1581.298091	42.23330557	39.4240781	34.7571709

3. Before performing the analysis, the user is asked to provide a range for the longitudinal area and the ties. The program uses these values to perform the following steps:
 - First finds the minimum and maximum reinforcement area out of all the beams in the structure.
 - Then it creates a list with possible reinforcement areas. The first item on the list is the range assigned by the user. Then the possible reinforcement groups increase

with step equal to the one assigned by the user until the value reaches the maximum required reinforcement.

Example:

Max reinforcement required = 800mm^2

Range = 200mm^2

Group	As
A	200
B	400
C	600
D	800

- Then the program compares the reinforcement of each member with the different reinforcement groups and assigns the reinforcement that is equal or slightly greater than the required one.
 - The same concept is applied for the top, bottom and ties.
 - Note that the possible groups for the longitudinal reinforcement are the same for top and bottom reinforcement, but the groups of the ties are a different group.
 - If the user provides a large range this will result in fewer groups, but a lot of the members will be excessively reinforced. If the user selects a small range this will result in a large number of groups, but the reinforcement provided will be almost equal to the required (provided by etabs).
 - The groups are shown in the output excel file. Longitudinal Groups and Ties Groups
4. The results of the previous steps can be found in the sheet Final Table – Groups of the output file. The column Group Name combines the name of the cross section, the top rebar group, the bottom rebar group and the tie group.

Section_Top_Bottom_Ties

	A	B	C	D	E	F	G	H	I
	DesignSect	UniqueName	As Top Total (mm ²)	As Bottom Total (mm ²)	As Transverse Total (mm ² /m)	Top Reinforcement Groups	Bottom Reinforcement Groups	Ties Reinforcement Groups	Group Name
1	Beam 30x50	100	712	718	955.05 A	A	A	B	Beam 30x50 - A_A_B
2	Beam 30x60	1001	1548	1233	2081.01 B	B	B	C	Beam 30x60 - B_B_C
3	Beam 30x60	1002	1976	909	2236.01 C	B	C	C	Beam 30x60 - C_B_C
4	Beam 30x60	1003	1293	760	1383.41 B	A	B	B	Beam 30x60 - B_A_B
5	Beam 30x60	1004	4004	2427	3025.93 E	C	C	D	Beam 30x60 - E_C_D
6	Beam 30x60	1005	3818	2233	3315.36 E	C	D	D	Beam 30x60 - E_C_D
7	Beam 30x60	1006	3048	2287	3299.2 D	C	C	D	Beam 30x60 - D_C_D
8	Beam 30x60	1007	3909	2247	3929.32 E	C	C	E	Beam 30x60 - E_C_E
9	Beam 30x60	1008	2282	1507	2173.34 C	B	C	C	Beam 30x60 - C_B_C
10	Beam 30x60	1009	3212	1686	2653.55 D	B	C	C	Beam 30x60 - D_B_C
11	Beam 30x50	101	1590	973	1862.48 B	B	C	C	Beam 30x50 - B_B_C
12	Beam 30x60	1010	1413	760	1387.61 B	A	B	B	Beam 30x60 - B_A_B
13	Beam 40x60	1011	5135	2792	4554.73 F	D	D	E	Beam 40x60 - F_D_E
14	Beam 30x60	1012	3115	2066	3160.98 D	C	C	D	Beam 30x60 - D_C_D
15	Beam 30x60	1013	3581	2109	2797.17 D	C	D	D	Beam 30x60 - D_C_D
16	Beam 80x70	1014	3629	2402	3983.79 D	C	C	E	Beam 80x70 - D_C_E
17	Beam 30x60	1015	1186	760	1666.69 B	A	B	B	Beam 30x60 - B_A_B
18	Beam 30x60	1016	1755	930	2175.96 B	B	C	C	Beam 30x60 - B_B_C
19	Beam 30x60	1017	4023	2577	4350.31 E	C	C	E	Beam 30x60 - E_C_E
20	Beam 30x60	1018	2007	1768	2088.35 D	B	D	D	Beam 30x60 - D_B_D

If the structure has a lot of beams this solution will result in a lot of different groups

5. Sheet Final Table Simplified – Groups of the output file, groups the longitudinal are into one group. For each beam it compares which face of the beam belongs to the group with the biggest reinforcement area and provides the same area (group) to the other face of the beam. Therefore, the Group Name column has the following form

Section_LongitudinalBars_Ties

Note that if a beam is assigned to group A for longitudinal bars, this means that A (Group area) must be placed at the top and A (Group area) at the bottom of the beam.

6. Finally sheet Summary Count of the output file shows the number of groups created with the two methods. Final Table – Groups and Final Table Simplified – Groups.

Description	Count
Unique Group Names (Simplified Table)	136
Unique Group Names (Full Table)	206

The analysis begins when the user presses the Run button

Once the message Done is shown on the window the user is given the opportunity to save the output file wherever they want by pressing the Save Excel button

The screenshot shows the 'Beam Organizer' application window. At the top, it displays 'Input Excel: Mixed Use - Beams 24.10.25.xlsx' and a 'Load Excel...' button. Below this, there are two input fields: 'Range Longitudinal Area (mm²):' with the value '500' and 'Range Ties Area (mm²/m):' with the value '1570'. Under these fields are two buttons: 'Run' and 'Save Excel...'. A 'Log:' window is open, showing the following text: 'Running... Done. Sheets ready: Design Section Summary, Unique Names R einforcement, Longitudinal Groups, Ties Groups, Final Tabl e - Groups, Final Table Simplified - Groups, Summary Count s'. At the bottom of the window, it says 'Developed by KT'.